AY 128: Astronomy Data Science Lab

UC Berkeley, Spring 2026

This course consists of three data-centric laboratory experiments that draw on a variety of tools used by professional astronomers. Students will learn to procure and clean data (drawn from a variety of world-class astronomical facilities), assess the fidelity/quality of data, build and apply models to describe data, learn statistical and computational techniques to analyze data (e.g., Bayesian inference, machine learning, parallel computing), and effectively communicate data and associated scientific results. This class will make use of data from facilities such as *Kepler*, *Gaia*, the *Sloan Digital Sky Survey*, and the *Hubble Space Telescope* to explore the structure and composition of the Milky Way, stars, and galaxies throughout the local and distant Universe. There is a heavy emphasis on software development in the Python language, statistical techniques, and high-quality communication (e.g., written reports, oral presentations, and data visualization).

Classroom: In person Tuesday and Thursday 2:00 PM to 3:30 PM, 131 Campbell. Attendance and engagement with lectures is important. Please have your computer ready for interactive demos and to work on your labs in class.

Website: https://ucb-datalab.github.io

Office Hours: see course website.

Labs: This class features 3 labs plus one warm-up activity (Lab 0). We will introduce the topic and contents of each lab throughout the semester. You will generally have 3-4 weeks to complete each lab and write up a report. There are weekly checkpoints in each lab. Labs and checkpoints will be submitted via Gradescope.

Quizzes and Readings: There are no formal exams for this class.

Communication: Our preferred method of communication is through the class Ed Discussion site. You will receive a link to sign up for our channel. Email is OK for logistical items (e.g., scheduling a meeting time with an instructor), but Ed Discussion is preferred.

Recommended Textbooks:

"Statistics, Data Mining, and Machine Learning in Astronomy: A Practical Python Guide for the Analysis of Survey Data" by Željko Ivezić et al.

"An Introduction to Modern Astrophysics" by Bradley W. Carroll & Dale A. Ostlie

"Foundations of Astrophysics" by Barbara Ryden & Bradley Peterson

and select readings to be provided.

Prerequisites:

- This class assumes that you have completed introductory astrophysical instruction (Astro 7A and 7B, or higher such as Astro 160/161) as well as knowledge of calculus (including Math 53) and linear algebra (Math 54 or Physics 89)
- You should have proficiency or fluency in the Python programming language. This class heavily emphasizes software development, and is **not** the place to learn Python for the first time.

Lab Reports (60% of total grade):

- Submitted as lab reports along with a polished, executable Jupyter Notebook.
- due **before** specified due date/time.
- Late Labs & “Slip days”:
  - Each person is allocated 8 slip days for the semester.
    - These are days you can use to turn in a lab report late without any penalty by submitting the [Slip Day Form] which can be found in a pinned Ed post. **You must submit your slip day form no more than 24 hours after you turn in the assignment.** You can allocate them as you see fit throughout the semester, though **we will not allow changes to the number of slip days applied once grades for that lab are posted**. If you want to change the number of days you decide to use before the lab grades are posted, resubmit the Slip Day Form and the most recent submission will be used.
    - **NO** extensions for labs will be granted beyond the 8 slip days except for demonstrated extenuating circumstances.
    - Once all slip days are used, then late lab grading policy is -10% x raw score x number of days late. For example, a lab submitted 2 days late that got a raw score of 90 points would be reduced by -0.10 x 90 x 2 = -18 pt, yielding a total of 72 points for the lab.
    - Each unused slip day at the end of the semester is worth 0.25% extra credit toward your final class grade.
- Collaborate (talk, draw pictures, analyze data) with your lab mates, but you **MUST** implement separately (your own equations, code, plots, writing)
- *Lab 0: Introduction to ADQL and Gaia Data* [50 points]
- *Lab 1: Gaia, RR Lyrae stars, and Galactic Dust* [100 points]
- *Lab 2: Modeling Stellar Spectra* [100 points]
- *Lab 3: Galaxy Morphology with Convolutional Neural Networks* [100 points]

Lab Checkpoints (25% of total grade):

- Short assessments to gauge your understanding and completion of individual checkpoints to be submitted on Gradescope. Graded on completion though *all questions must be appropriately answered and some effort towards completion must be shown.* (**10% of total grade**)
- Group presentations to be given during class in groups of 3-5 members. Details on the format and expectations will be given during class. (**20% of total grade**)

Class Participation (15% of total grade):

- Participation can take many forms including asking questions during lecture, leading group discussion, participating in presentations, posting questions AND answers to others questions on Ed Discussion, attending office hours, and more.
- The key is for you to be actively engaged in the class.
- Historically, those with low class participation scores are unable to get an A in this class

Reading: Lab instructions and topical handouts linked on the class webpage

Materials: Each student will be given access to Datahub (https://astro.datahub.berkeley.edu) and the Savio HPC cluster (https://docs-research-it.berkeley.edu/services/high-performance-computing/overview/) to execute their code.

Class Philosophy: This class is effectively an introduction to astronomical research from a data-centric perspective. The labs are designed to be challenging, yet manageable mini-research projects. They will be time-consuming.
Often when doing research, you will not have all the necessary technical skills to complete a project. It is normal to have to learn new skills along the way. A main aim of this class is to provide exposure to those

skills, in a setting that requires you to think and act like a researcher, but with extra guidance and resources. You should also get used to asking questions that seem basic (sometimes embarrassingly so) in hindsight, and working in open and collaborative environments.

The weekly lectures will provide you with broad background, but are not intended to teach all the nitty-gritty technical skills you need. Much like real research, these are skills you learn on your own, in groups, and/or in consultation with an advisor. Ask questions via Ed Discussion and office hours, particularly with the GSIs.

Research is fun, but challenging. It usually does not progress linearly, and you often have to take a few steps backward before ultimately moving forward.

Class Conduct: *This is a work-intensive class.* You are going to spend significant time on your own in the lab with minimal supervision. At all times, you are expected to abide by the UC Berkeley Code of Conduct (https://sa.berkeley.edu/code-of-conduct), acting with respect to your peers, GSIs, and instructor. Should you experience any form of harassment or discrimination, we maintain a list of resources that can help you decide how to respond. (https://astro.berkeley.edu/department-resources/reporting-harassment). GSIs and instructors are non-confidential reporters; we have a legal obligation to act on any reports of harassment. Please know that we take our responsibility seriously.

Collaboration Policy (don't cheat): The Data Lab is a collaborative class. To repeat what is written previously in the syllabus: while you should talk with others about the labs and we strongly encourage collaboration, **we require that you write your solutions and reports individually.**

Use of AI: You are not allowed to use AI (e.g., Gemini, ChatGPT) to generate **submitted** material (i.e. checkpoints and labs) for this class. Such use of AI will result in a zero and, depending on the situation, may also be considered grounds for academic misconduct. You may use AI in other ways outlined in the department AI use policy, which you can find here. **You must make an "AI declaration of use" statement at the end of your lab report outlining in detail what tools were used and how to answer specific questions.** When in doubt, ask for clarification from the instructors.

DSP Accommodations: UC Berkeley is committed to creating a learning environment that meets the needs of its diverse student body including students with disabilities. If you anticipate or experience any barriers to learning in this course, please feel welcome to discuss your concerns with me.

If you have a disability, or think you may have a disability, you can work with the Disabled Students' Program (DSP) to request an official accommodation. The Disabled Students' Program (DSP) is the campus office responsible for authorizing disability-related academic accommodations, in cooperation with the students themselves and their instructors. You can find more information about DSP, including contact information and the application process here: dsp.berkeley.edu. If you have already been approved for accommodations through DSP, please meet with the course instructors so we can develop an implementation plan together. Note that per University policy, we can only provide DSP accommodations if an official letter is provided by the DSP office.

Waitlisted Students: If you are on the waiting list, you must still do all coursework and complete labs and homework by the deadlines. We will not be offering extensions if you are admitted into the course later. So it is your responsibility to stay up to date on the assignments.

You are welcome to attend lectures while you are on the waiting list. Rooms may feel a little crowded in the first week.

Unfortunately, doing all the work is not a guarantee of enrollment. You will only be enrolled if there is space in your lab. Enrollment will proceed by CalCentral.

Berkeley Astro Technology Access: We are committed to ensuring that every student in our department is able to engage in our courses without incurring undue financial burdens.

If you are in need of a personal laptop or other technology to complete academic work this semester, the Student Technology Equity Program (STEP) can provide need-based loans for a variety of hardware to Berkeley students. Applications are now open for the Spring 2026 semester: https://studenttech.berkeley.edu/step. We encourage you to apply as soon as possible as it will take 1-2 weeks for the application to be processed.

If you are in need of shorter term access to technology, Berkeley Library offers first come, first serve laptop lending. These loans are only available for up to 14 days, so they are not a solution for a full semester, but can help for moments where you may need short term access to a computer, including the waiting period before your STEP loan is approved. See here for the checkout form and a complete list of offerings: https://www.lib.berkeley.edu/about/device-lending-policy.

Your success!
We want you to succeed in this class. We know that Berkeley can at times be an overwhelming place. If you find yourself struggling to keep up, please come talk to us, we're here to help you succeed.